

1. Polynomial Spaces

Let $\mathcal{P}_n$ denote a real-valued vector space of polynomials of degree less than or equal to n , where n is a nonnegative integer. A generic polynomial in $\mathcal{P}_n$ can be expressed as follows:

$$p(t) = \sum_{k=0}^n p_k t^k = \underbrace{\begin{bmatrix} 1 & t & \cdots & t^n \end{bmatrix}}_{\mathbf{f}^\top(t)} \underbrace{\begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_n \end{bmatrix}}_{\mathbf{p}} = \mathbf{f}^\top(t) \mathbf{p},$$

where $t \in \mathbb{R}$, $\mathbf{f}(t) \in \mathbb{R}^{n+1}$ denotes the vector of monomials (you can think of it as a vector-valued function of t), $\mathbf{p} \in \mathbb{R}^{n+1}$ denotes the vector of the coefficients, and $^\top$ denotes the transpose.

Define $\mathbb{V} \subseteq \mathcal{P}_n$ as the subset of all polynomials in $\mathcal{P}_n$ that have $t = 0$ as a root. That is,

$$\mathbb{V} = \left\{ p(t) = \sum_{k=0}^n p_k t^k \mid p(0) = 0, p_k \in \mathbb{R}, k = 0, \dots, n \right\}.$$

- Explain why $\mathbb{V}$ is a subspace of $\mathcal{P}_n$.
- Show that each polynomial in $\mathcal{P}_n$ can be represented by a unique vector in $\mathbb{R}^{n+1}$. Similarly, show that each polynomial in $\mathbb{V}$ can be represented by a unique vector in $\mathbb{R}^n$.

2. Fun with Cauchy-Schwarz

The Cauchy-Schwarz inequality states that for two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| = |\mathbf{x}^\top \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

(a) Without using a calculator, show that

$$1(100) + 2(99) + 3(98) + \dots + 100(1) \leq 1^2 + 2^2 + 3^2 + \dots + 100^2$$

(b) Show that for any $\mathbf{v} \in \mathbb{R}^n$ such that all entries $v_i \geq 0$

$$\|\mathbf{v}\|_1 \leq \sqrt{n} \|\mathbf{v}\|_2$$

HINT: Recall the following definition of the 1-norm, $\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$